BANKER'S ALGORITHM

CODE

Activities Terminal Apr 20 08:29

tejini@localhost:~/scheduling — /usr/bin/vim banker.c

```
        }
        while(count<n)
        {
                safeFound=0;
                for(i=0;i<n;i++){
                        if(finish[i]==0)
                        {
                                int safe=1;
                                for(j=0;j<r;j++){
                                        if(need[i][j]>avail[j]){
                                                safe=0;
                                                break;
                                        }
                                }
                                if(safe){
                                        for(k=0;k<r;k++){
                                                avail[k]+=alloc[i][k];
                                        }
                                        safe_sequence[count]=i;
                                        finish[i]=1;
                                        safeFound=1;
                                        count++;
                                }
                        }
                }
                if(!safeFound){
                        printf("System is in unsafe state! May occur deadlock.\n");
                        return 0;
                }
        }

        printf("safe sequence is: ");
        for(i=0;i<n;i++){
                printf("P%d ",safe_sequence[i]);
        }
        printf("\n");
        return 0;
}
                                                                        81,1-8        Bot
```

Activities Terminal Apr 20 08:28

tejini@localhost:~/scheduling — /usr/bin/vim banker.c

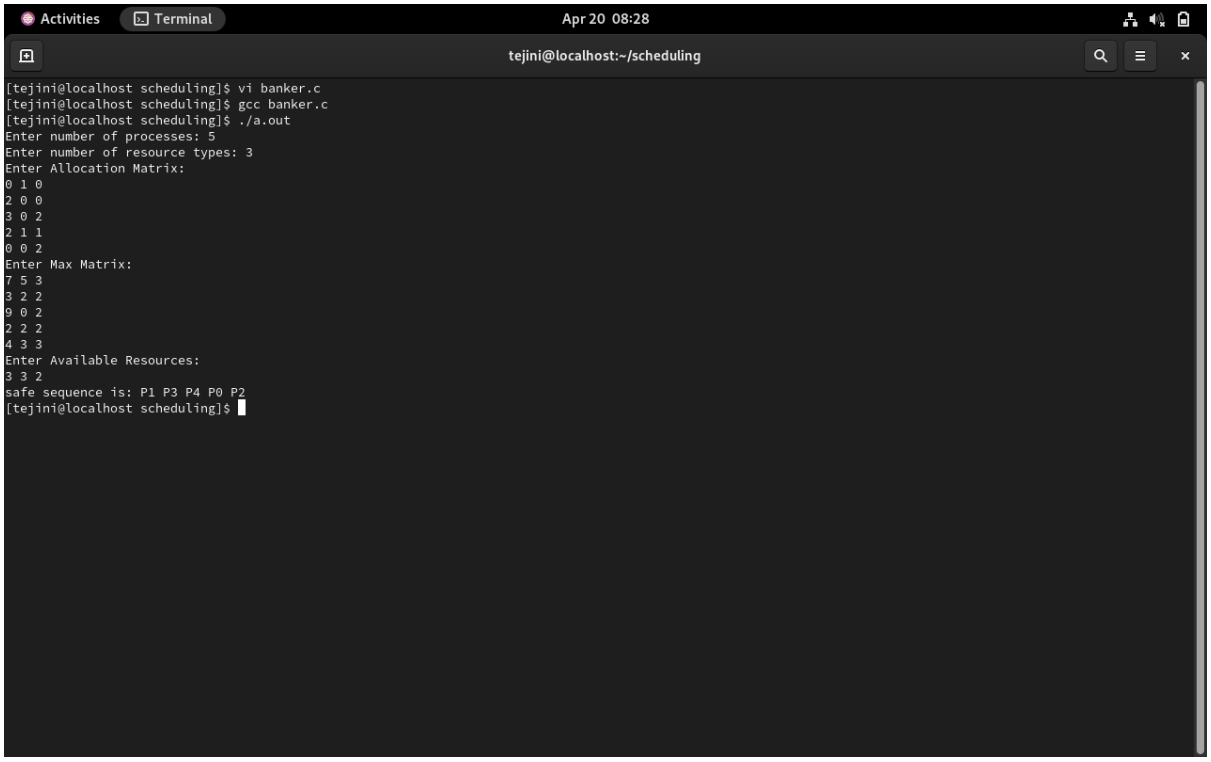
```
#include<stdio.h>
int main(){
        int n,r,i,j,k;
        printf("Enter number of processes: ");
        scanf("%d",&n);
        printf("Enter number of resource types: ");
        scanf("%d", &r);
        int alloc[n][r],max[n][r],need[n][r],avail[r];
        int finish[n],safe_sequence[n];
        int count = 0, safeFound;

        printf("Enter Allocation Matrix:\n");
        for(i=0;i<n;i++)
        {
                for(j=0;j<r;j++)
                {
                        scanf("%d",&alloc[i][j]);
                }
        }

        printf("Enter Max Matrix:\n");
        for(i=0;i<n;i++){
                for(j=0;j<r;j++)
                {
                        scanf("%d",&max[i][j]);
                }
        }

        printf("Enter Available Resources:\n");
        for(i=0;i<r;i++){
                scanf("%d",&avail[i]);
        }
        for(int i=0;i<n;i++)
        {
                finish[i]=0;
        }
        for(i=0;i<n;i++)
        {
                for(j=0;j<r;j++){
                        need[i][j]=max[i][j]-alloc[i][j];
                }
"banker.c" 82L, 1298B                                                   1,17          Top
```

OUTPUT :



```
[tejini@localhost scheduling]$ vi banker.c
[tejini@localhost scheduling]$ gcc banker.c
[tejini@localhost scheduling]$ ./a.out
Enter number of processes: 5
Enter number of resource types: 3
Enter Allocation Matrix:
0 1 0
2 0 0
3 0 2
2 1 1
0 0 2
Enter Max Matrix:
7 5 3
3 2 2
9 0 2
2 2 2
4 3 3
Enter Available Resources:
3 3 2
safe sequence is: P1 P3 P4 P0 P2
[tejini@localhost scheduling]$
```